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原生恆星盤外圍絲狀結構之模型與觀測比較
Modeling Streamer Structures in Collapsing
Prestellar Cores

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摘要

Streamers have been observed with high-resolution ALMA observations around many protoplanetary disks undergoing formation. They have been suggested to dominate the mass accretion budget from the collapsing prestellar core, and have important consequences on the disk dynamics. Understanding the formation of streamers is therefore important for explaining how protoplanetary disks receive mass from the envelope. We propose a model to explain the formation of streamer structures by considering density enhancements due to gravitational instability. We test our model against sources where streamers have been detected (ex., Per-emb-2 and Per-emb-50 observed with NOEMA) and fit for model parameters. This allows us to gain deeper insights into the physical origin of streamers and their role in mass transport from the core to the disk. This model can be applied to analyze new observations as well as archival data that show signs of streamers.

Abstract

Streamers have been observed with high-resolution ALMA observations around many protoplanetary disks undergoing formation. They have been suggested to dominate the mass accretion budget from the collapsing prestellar core, and have important consequences on the disk dynamics. Understanding the formation of streamers is therefore important for explaining how protoplanetary disks receive mass from the envelope. We propose a model to explain the formation of streamer structures by considering density enhancements due to gravitational instability. We test our model against sources where streamers have been detected (ex., Per-emb-2 and Per-emb-50 observed with NOEMA) and fit for model parameters. This allows us to gain deeper insights into the physical origin of streamers and their role in mass transport from the core to the disk. This model can be applied to analyze new observations as well as archival data that show signs of streamers.

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Chapter 1

Introduction

1.1 The formation of cores

In a molecular cloud, a local density enhancement can trigger gravitational collapse, forming a prestellar core. As the collapse proceeds, the prestellar core becomes a protostellar core. During the early class 0 or class I stage, these systems are still deeply embedded within their envelope. Within this turbulent environment, small fluctuations can lead to gravitational instability, giving rise to an asymmetric, filamentary structure known as the streamer. Streamer is the density enhancement structures within the protostellar envelope. The key characteristics are that do not necessarily follow the Keplerian rotation. They have been considered as a channel to carry mass onto the disk.

1.2 The streamer in observations

Recently, several observations revealed asymmetric structures in the protostellar system (e.g., Pineda et al. 2020, Valdivia-Mena et al. 2022, Hsieh et al. 2023). The features of streamers are very diverse in morphology, do not always form in a disk plane, and have very weak density enhancement in observation.

1.3 Observations

To test our model against sources where streamers have been detected, we use five observational datasets: Per-emb-2, Per-emb-50, S CrA, HL Tau, and SVS 13A. The Per-emb-2, Per-emb-50, and SVS 13A were observed by the Northern Extended Millimeter Array (NOEMA). Per-emb-2 was observed in D-configuration on June 22, July 27-28, and August 19, 2018, and in C-configuration on October 9, 10, 17, and 18 (project S18AG, Pineda et al. 2020). The baseline values ranged from 16.4 to 287.8 meters. Per-emb-50 was observed in C-configuration on December 29, 2019, and January 5, 2021, and D-configuration on August 6, September 7, and 8, 2020 (Valdivia-Mena et al. 2022). SVS13A was also observed using two different configurations: C-configuration on December 29, 2019, and January 5, 2020, and in D-configuration on August 6 and September 7, 2020. The baseline range is between 15.6 and 368 meters, corresponding to a synthesized beam size of $1.''2 \times 0.''7$ (Hsieh et al. 2023). The S CrA , HL Tau were observed by Atacama Large Millimeter/submillimeter Array (ALMA). The S CrA observation data from ALMA Science Archive was observed on November 6, 2019 (Project 2019.1.01792.S, Gupta et al. 2023). HL Tau was observed by ALMA’s C40-7 array configuration on August 22, 28, and September 1, 2017. The maximum projected baseline length is 3.3 km (Yen et al. 2019). For each source, we extract the position-position-velocity (PPV) data of the streamer.

1.3.1 Per-emb-2

1.3.2 Per-emb-50

1.3.3 SVS 13A

1.3.4 S CrA

1.3.5 HL Tau

1.4 Our work overview

The formation process of protoplanetary disks is still unrefined. Given their importance, we have developed a new model to characterize the streamer features and kinematics, focusing on gravitationally driven infalling and rotation.

Chapter 2

Methods

2.1 Build up a model

We build a model that can reproduce the observed streamer by considering a self-collapse density enhancement structure with a rotation velocity and evolve with time in 3D space.

2.2 Formalism

In the formalism, including the density enhancement along the polar angle (θ), infall velocity (v_{inf}), rotation velocity (v_{rot}), accumulated rotation (ϕ), and inclination (i). The sonic length.

$$r_s = c_s T \quad (2.1)$$

Infall velocity

$$v_{\text{inf}} = -\sqrt{\frac{2GM_*}{r}} - 3.3c_s \quad (c_s = 200m/s) \quad (2.2)$$

The rotation velocity.

$$v_{\text{rot}} = r_s \omega \Omega_{\text{kep}} \left[\left(\frac{r}{r_s} \right)^{-1} + \left(\frac{r}{r_s} \right)^{-\frac{1}{3}} \right] \quad (2.3)$$

The accumulated rotation.

$$\phi = \phi_0 + t_s \omega \Omega_{\text{kep}} \sqrt{\frac{2c_s^3 t_s}{GM_*}} \left[\left(\frac{r}{r_s} \right)^{-\frac{1}{2}} + \left(\frac{r}{r_s} \right)^{-\frac{1}{3}} \right] \quad (2.4)$$

2.3 Coordinate transformation

Position

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases} \quad (2.5)$$

Infall velocity

$$\begin{cases} v_{x_{\text{inf}}} = v_{\text{inf}} \sin \theta \cos \phi \\ v_{y_{\text{inf}}} = v_{\text{inf}} \sin \theta \sin \phi \\ v_{z_{\text{inf}}} = v_{\text{inf}} \cos \theta \end{cases} \quad (2.6)$$

Rotation velocity

$$\begin{cases} v_x = -v_{\text{rot}} \sin \theta \sin \phi + v_{x_{\text{inf}}} \\ v_y = v_{\text{rot}} \sin \theta \cos \phi + v_{y_{\text{inf}}} \\ v_z = 0 + v_{z_{\text{inf}}} \end{cases} \quad (2.7)$$

2.4 Streamer on the sky in the PPV space

Position rotates by inclination. Inclination(i): backward (+), forward (-).

$$\begin{cases} x' = r \sin \theta \cos \phi \\ y' = y \cos i + z \sin i \\ z' = -y \sin i + z \cos i = r(-\sin \theta \sin \phi \sin i + \cos \theta \cos i) \end{cases} \quad (2.8)$$

Velocity rotates by inclination

$$\begin{cases} v'_x = v_x \\ v'_y = v_y \cos i + v_z \sin i = v_{rot} \sin \theta \cos \phi \cos i + v_{inf}(\sin \theta \sin \phi \cos i + \cos \theta \sin i) \\ v'_z = -v_y \sin i + v_z \cos i = -v_{rot} \sin \theta \cos \phi \sin i + v_{inf}(-\sin \theta \sin \phi \sin i + \cos \theta \cos i) \end{cases} \quad (2.9)$$

θ corresponds to the initial polar angle of the streamer, ϕ is the initial azimuth angle, i is the inclination angle (zero inclination corresponds to an edge-on disk), ω is the ratio of Keplerian rotation velocity, and t_s is the collapsing time of the particle within the streamer.

2.5 Fitting procedure

The fitting procedure described below is applied to each dataset to constrain the model parameters. The results are compared to the observed streamer morphology and kinematics to assess the validity of the model.

2.5.1 Position angle

[(YN) This step can be neglected if the disk PA on the sky is known. Otherwise, the previous step needs to be repeated until the optimal is

found.]

2.5.2 Global fitting of the full model

2.6 Fitting with observations

2.6.1 Fitting with partially known parameters

Chapter 3

Results

Chapter 4

Discussion

Chapter 5

Conclusion

Appendix A

Observations

Table A.1: Observations

Target	RA (J2000) (h m s)	Decl (J2000) (° ' ")	Stellar mass ($M_{\odot}$)	Systemic velocity (km/s)	Disk inclination	Disk position angle	Reference
Per-emb-2	03 : 32 : 17.9320	30.49.47.7050	3.2				1
Per-emb-50	03 : 29 : 07.7680	31.21.57.1280	2.58		67°	170°	2
HLTau	04 : 31 : 38.4280	18.13.57.1900	2.1	7.14	47°	°	3
S CrA	19 : 01 : 08.6000	−36.57.20.0000	2	5.86	65°	°	4
SVS 13A	03 : 29 : 03.7400	31.16.03.8000			°	°	5

Note The inclination angle with $i=0$ corresponds to a face-on disk. **References**

⁽¹⁾Pineda et al. 2020 ⁽²⁾Valdivia-Mena et al. 2022 ⁽³⁾Yen et al. 2019 ⁽⁴⁾Gahm et al. 2018 ⁽⁵⁾Hsieh et al. 2023

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